DATABASE MANAGEMENT LAB Report

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

to

K.R Mangalam University

Submitted by:

RADHIKA

(2401010094)

Submitted to

Ms. Mansi Kajal

(Assistant Professor)

Department of Computer Science and Engineering School of Engineering and Technology
K.R Mangalam University, Gurugram- 122001, India SESSION – 2025-2026

EXPERIMENT - 1

Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:

COLLEGE DATABASE:

STUDENT (USN, SName, Address, Phone, Gender)

SEMSEC (SSID, Sem, Sec)

CLASS (USN, SSID)

SUBJECT (Subcode, Title, Sem, Credits)

IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

Solution:

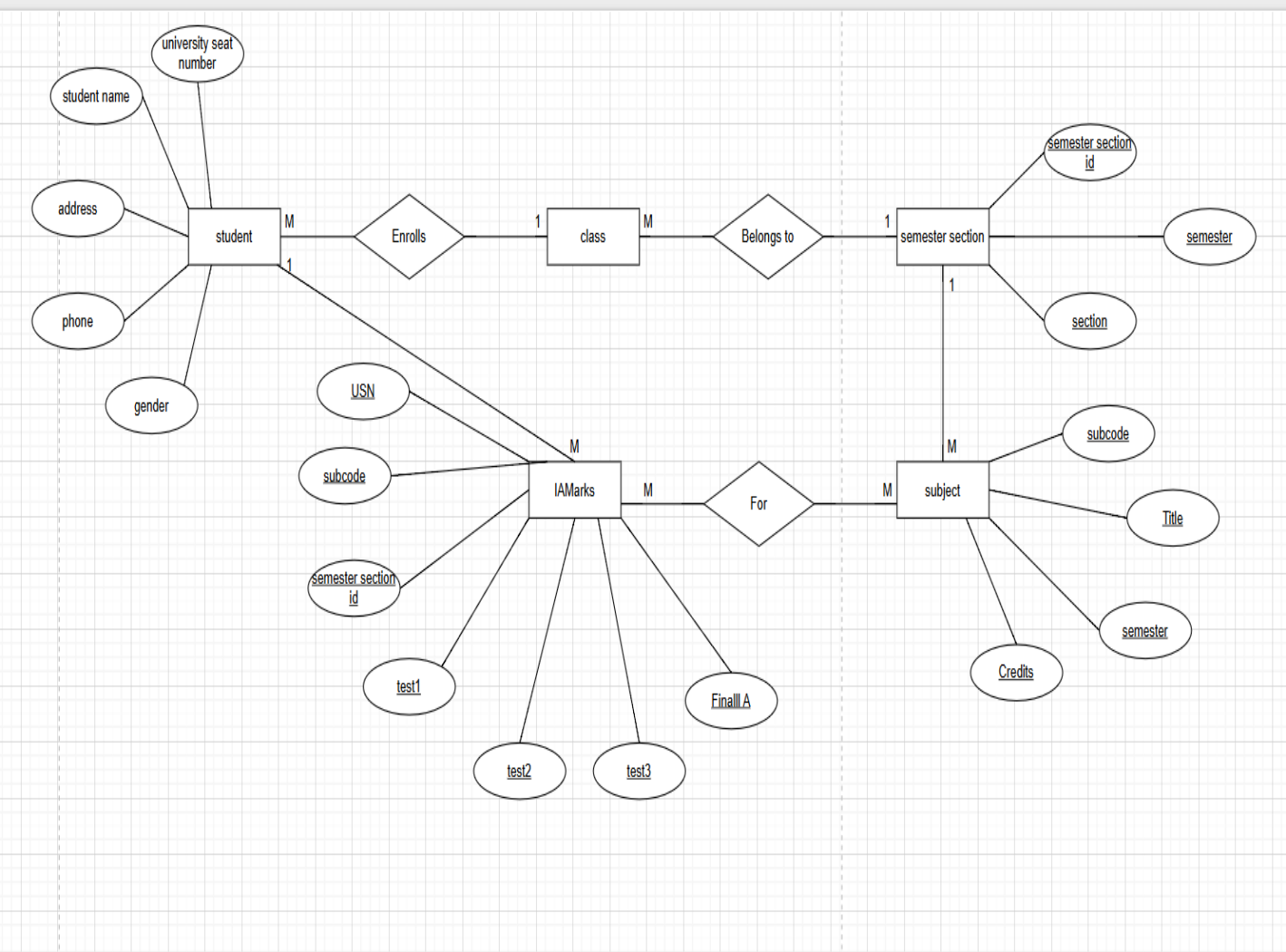
1. The strong entities STUDENT, SEMSEC, and SUBJECT are identified and represented using rectangles.
2. The associative entity IAMARKS is identified to resolve the M : N relationship between STUDENT and SUBJECT and is also represented using a rectangle.
3. The attributes are represented using ellipses:

- STUDENT: *USN* (Primary Key), SName, Address, Phone No, Gender
  Phone No is shown using a double ellipse as it is a multivalued attribute.
- SEMSEC: *SSID* (Primary Key), Sem, Sec
- SUBJECT: *Subcode* (Primary Key), Title, Sem, Credits
- IAMARKS: USN, Subcode, SSID, Test1, Test2, Test3, FinalIA

4. Primary keys are represented by underlining the attribute names.

5. The composite primary key of IAMARKS is (USN, Subcode, SSID).
6. CLASS is drawn as a relationship (diamond) between STUDENT and SEMSEC with M : N cardinality.
7. The relationship between STUDENT and IAMARKS is 1 : M, indicating that one student can have many IA records.
8. The relationship between SUBJECT and IAMARKS is 1 : M, indicating that one subject can have many IA records.
9. The relationship between SEMSEC and IAMARKS is 1 : M, indicating that marks are recorded semester-section wise.
10. All relationships, attributes, primary keys, and cardinalities are clearly marked in the ER diagram.

ER DAIGRAM:



EXPERIMENT-2

Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:

COMPANY DATABASE:

EMPLOYEE (SSN, Name, Address, Sex, Salary, SuperSSN, DNo)

DEPARTMENT (DNo, DName, MgrSSN, MgrStartDate)

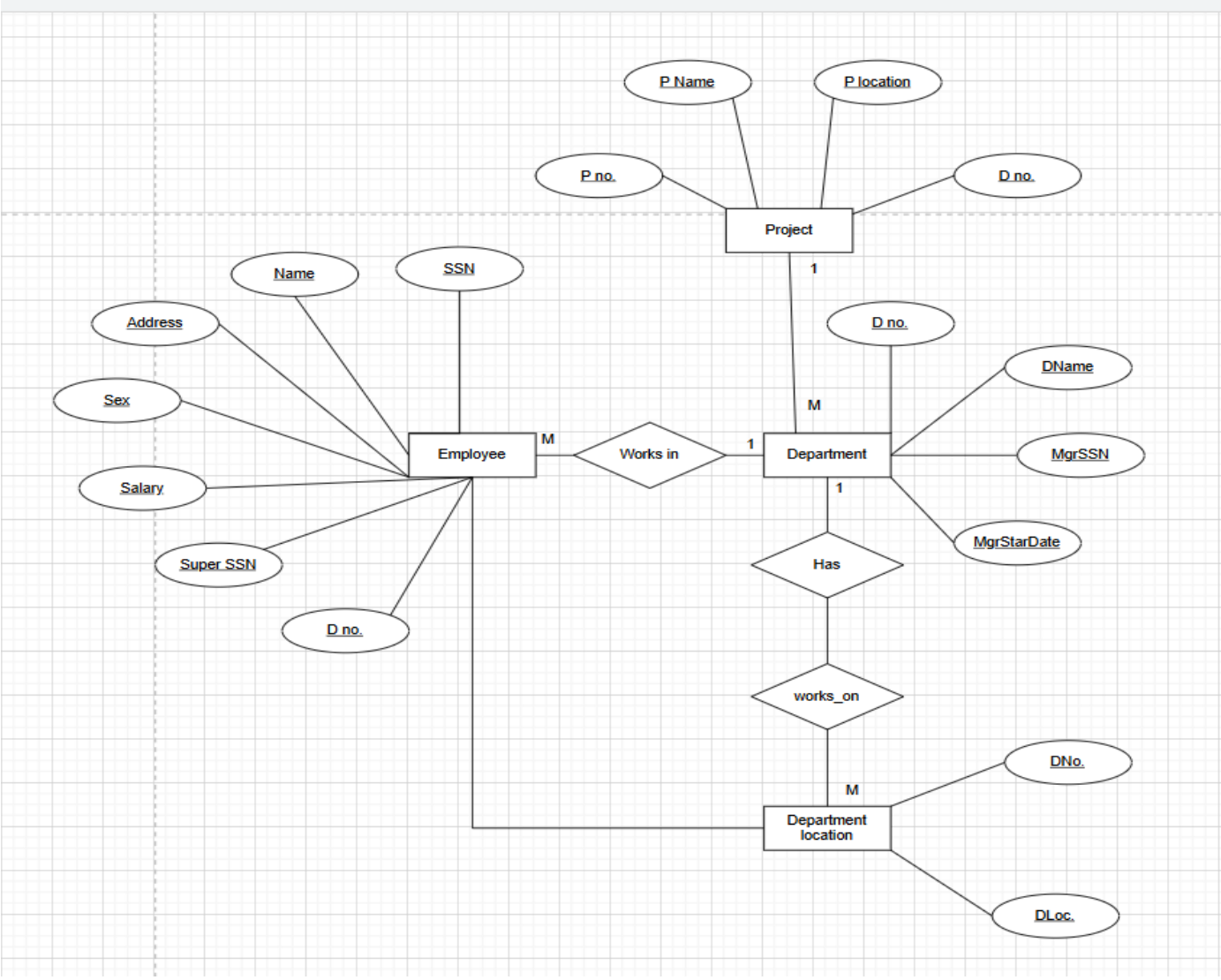
DLOCATION (DNo,DLoc)

PROJECT (PNo, PName, PLocation, DNo)

WORKS_ON (SSN, PNo, Hours)

SOLUTION:

1. Identify the entities EMPLOYEE, DEPARTMENT, PROJECT, and DLOCATION and represent them using rectangles in the ER diagram.
2. Add attributes using ovals and underline Primary Keys:
   - EMPLOYEE:
     SSN (Primary Key), Name, Address, Sex, Salary, SuperSSN, DNo
   - DEPARTMENT:
     DNo (Primary Key), DName, MgrSSN, MgrStartDate
   - PROJECT:
     PNo (Primary Key), PName, PLocation, DNo
   - DLOCATION:
     DNo, *DLoc* (Composite Primary Key)
3. Draw WORKS_ON as a diamond between EMPLOYEE and PROJECT with M : N cardinality and add attribute Hours to the diamond.
4. Draw MANAGES relationship between EMPLOYEE and DEPARTMENT with 1 : 1 cardinality and connect MgrSSN accordingly.
5. Connect EMPLOYEE to DEPARTMENT with 1 : M relationship (Many employees work in one department).

6. Connect DEPARTMENT to PROJECT with 1 : M relationship (One department controls many projects).
7. Connect DEPARTMENT to DLOCATION with 1 : M relationship (One department can have multiple locations).
8. Draw recursive relationship SUPERVISES between EMPLOYEE and EMPLOYEE with 1 : M cardinality (One supervisor manages many employees).
9. Clearly mark all cardinalities (1:1, 1:M, M:N) near relationship lines.
10. Underline all Primary Keys in each entity.

ER DAIGRAM: